Curriculum Vitae

Personal details

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Personal Statement

I am a Ph.D. student in Chemical and Biomolecular Engineering at KAIST, pursuing *material- and process-level engineering* of organic and inorganic semiconductors, and their device-level applications in neuromorphic systems, with a particular focus on *initiated Chemical Vapor Deposition (iCVD)*.

Academic Education

Mar 2025 – Current	Ph.D., Dept. of Chemical and Biomolecular Engineering at KAIST GPA: 4.03/4.3 Advisor: Prof. Sung Gap Im
Mar 2023 – Feb 2025	M.S., Dept. of Chemical and Biomolecular Engineering at KAIST GPA: 4.03/4.3 Advisor: Prof. Sung Gap Im
Mar 2019 – Feb 2023	B.S., Dept. of Chemical Engineering at Pohang University of Science and Technology (POSTECH) GPA: 4.06/4.3 (Summa Cum Laude)

Work Experiences

May 2026 – Nov 2026	Visiting Scholar, Dept. of Materials Science and Engineering at Pennsylvania State University (University Park, Supervisor: Prof. Saptarshi Das)
Jul 2022 – Aug 2022	Intern, LOTTE Chemical Intellectual Properties Strategy Team
Jan 2022 – Feb 2022	Intern, Korea Advanced Institute of Science and Technology (KAIST), Functional Thin Film Laboratory (Supervisor: Prof. Sung Gap Im)
Sep 2021 – Dec 2021	Intern, Pohang University of Science and Technology (POSTECH), Chemical Engineering Lab (Supervisor: Prof. Won Bae Kim)
Jun 2021 – Aug 2021	Intern, SK Hynix Future Technology R&D Center NAND Etch Team

Honors and Awards

Mar 2023 – Current	KAIST Government Scholarship (Full tuition)
Mar 2021 – Feb 2023	National Science & Technology Scholarship, Korea Student Aid Foundation (Full tuition, merit-based scholarship)
Mar 2019 – Feb 2021	Jigok Scholarship, POSTECH (Full tuition for 4 semesters)

Research Area

1. Development of p-type tellurium-based electronic devices

a) High-hole-mobility p-type tellurium thin-film transistors enabled by iCVD-based polymeric passivation layers

- Exploration and optimization of polymeric passivation layers to enhance switching characteristics of p-type transistor devices, including on/off ratio, carrier mobility, and threshold voltage
- Analysis of carrier transport mechanisms through spectroscopic techniques (e.g., UPS, Raman) and electrical characterization (e.g., TLM, Schottky barrier analysis)
- Development of vertically integrated CMOS logic circuits and flexible TFTs

b) Enhancement of contact properties in Te TFTs using thermally decomposable polymeric contact interlayers for an ash-out strategy

- Exploration and optimization of polymeric buffer layers to reduce contact resistance and enable higher current levels
- Investigation of the ash-out process using in-situ heating-cell ellipsometry

2. High-performance vertical organic thin-film transistor devices

a) Vertical organic thin-film transistors with vertically aligned carrier transport for high-current and high-frequency operation

- Design of unconventional vertical organic thin-film transistor architectures and circuit layouts for complex logic circuits (e.g., ring oscillators, half adders)
- Analysis of high-frequency device operation using a fast waveform generator and analyzer module (WGFMU) integrated into the B1500A

3. Next-generation neuromorphic devices and their applications

a) Gate-injection-type organic synaptic transistors with tunable decay times for reservoir networks targeting human activity recognition

- Rational selection and optimization of charge-storage-layer semiconductor materials to modulate gate-induced charge transfer and control EPSC decay characteristics
- Implementation of synaptic transistors in various AI and neuromorphic applications (e.g., NeuroSim, machine learning), with a particular focus on human activity recognition tasks using the UCI-HAR dataset

Research skills and experiences

Characterization Techniques

- SEM, AFM, FT-IR, XPS · depth profiling, XPS · high-resolution, UPS · depth profiling, Contact angle analyzer, Ellipsometer (optical analysis), XRD, Hall effect measurement

Polymer Fabrication & Surface Engineering

- iCVD-based functional homopolymer and copolymer synthesis, Composition and structure optimization

Fabrication and Characterization of Electronic Devices

- 2-terminal devices: MIM, RRAM, 3-terminal devices: TFT, flash memory, synaptic devices, Multi-terminal devices/circuits: inverter, logic gates, ring oscillator, 2D material-based transistor fabrication: sourcing, exfoliation, wet transfer, Device layout designing, Optoelectronic measurement

Devices & Instruments

- iCVD system, Atomic layer deposition (ALD) system, Metal/organic thermal evaporator, E-beam evaporator, E-beam lithography, Photo-lithography mask aligner, Reactive ion etching (RIE) system, Plasma treatment system, Optical microscope, Keysight B1500A semiconductor analyzer, Probe station in glove box with temperature-controlled chuck, Oscilloscope and function generator

Software & Computation

- Python (TensorFlow, Scikit-learn), C, C++, Java, MATLAB, Quantum Espresso, RASPA, ORCA, LAMMPS, OriginLab, ChemDraw, ImageJ, Raith lithography

Making & General

- Fusion360, Cura slicer, Homebrew 3D printer, WordPress, Apache server, MS Office, Linux, Photoshop, Illustrator, Blender

Publications (SCIE & ESCI)

1. **Taehyun Nam**†, Seung Min Lee†, Chungryeol Lee, Changhyeon Lee, Sunwoo Jeong, Seunghwan Seo, Youson Kim, Jeong-ik Park, Seokhyun Hong, Hyung Joong Yun, Kibum Kang, Hocheon Yoo, Junhwan Choi*, and Sung Gap Im*, A High-Hole Mobility Tellurium Transistor with Electron Donating Channel Passivation Layer for Scalable, High-Throughput Electronics, *Advanced Functional Materials*, **36**, e27125, (2026). doi:10.1002/adfm.202527125
2. Youson Kim†, **Taehyun Nam**†, Junyeong Yang, Changsoon Choi, Seun-Bae Jeon* and Sung Gap Im*, Electrolyte–channel interface engineering for STM/LTM differentiation in OECTs for physical reservoir computing, *Nature Communications*, In submission.
3. Junyeong Yang†, Sanghyun Kim†, Jeong-Ik Park†, Minjeong Kang, Gyuwon Lee, Changmin Lee, **Taehyun Nam**, and Sung Gap Im*, Sub-0.3 V Ultra-Low-Voltage Operation of Bilayer RRAM via High-Sulfur Polymer/Inorganic Hybrid Layers for Hardware Synaptic Devices, *Advanced Functional Materials*, In submission.
4. Chungryeol Lee†, Dongho Choi†, Sanghoon Park, Jeong-ik Park, **Taehyun Nam**, Minjae Jang, Seunghyup Yoo*, and Sung Gap Im*, A light-driven multi-state heterojunction transistor for optoelectronic ternary logic circuits, *Nature Communications*, Accepted, (2026).
5. Sukwon Jang†, Keunho Soh†, Chungryeol Lee†, **Taehyun Nam**, Minjae Jang, Jeong-ik Park, Changhyeon Lee, Junhwan Choi, Jung Ho Yoon*, and Sung Gap Im*, A Unipolar-Driven Synaptic Transistor for Environment-Adaptable Vision System, *Nature Communications*, **16**, 7636, (2025). doi:10.1038/s41467-025-63073-2
6. Sukwon Jang†, Youson Kim†, Chungryeol Lee†, **Taehyun Nam**, Jeong-ik Park, Junyeong Yang, Juchan Kim, Bohyun Lee, Sung Gap Im*, Vapor-Phase Deposited Polymer Dielectric Layers for Organic Electronics: Design, Characteristics, and Applications, *Korean Journal of Chemical Engineering*, **42**, 1931–1949, (2025). doi:10.1007/s11814-024-00210-5
7. Jeong-ik Park†, Changhyeon Lee†, Chungryeol Lee, **Taehyun Nam**, Yebin Bak, Sukwon Jang, Hocheon Yoo, Junhwan Choi*, and Sung Gap Im*, Dynamically Reconfigurable Multibit Modulator Enabled by Programmable Anti-Ambipolar Memory Transistors, *Advanced Functional Materials*, In revision.
8. **Taehyun Nam**†, Chungryeol Lee†, Seokhyun Hong, Jeong-ik Park, Youson Kim, and Sung Gap Im*, Gate-Injection Synaptic Transistor with Tunable Stochastic Memory for Ensemble Reservoir Computing, In preparation.
9. Seokhyun Hong†, **Taehyun Nam**†, Chungryeol Lee†, Jeong-ik Park, Youson Kim, Junhwan Choi* and Sung Gap Im*, High-current Organic Vertical Transistor with Bottom-up Grain Boundary Selective Deposition of Functional Polymeric Layer, In preparation.

(† These authors contributed equally to this work. * Corresponding author.)

Conferences and Presentations

1. **Taehyun Nam**, Seung Min Lee, Junhwan Choi*, Sung Gap Im*, “High-Performance p-Type Tel-

lurium Transistor Arrays and Circuits via Vapor Phase Deposited Polymer Passivation", The 24th International Meeting on Information Display (IMID 2024), August 21, 2024, Jeju, South Korea

2. **Taehyun Nam**, Seung Min Lee, Junhwan Choi*, Sung Gap Im*, "A High-Hole Mobility Tellurium Transistor with Electron Donating Channel Passivation Layer for Scalable, High-Throughput Electronics", The 100th Annual Spring Meeting of the Polymer Society of Korea, April 9, 2026, Daejeon, South Korea

Patents

1. Sung Gap Im, **Taehyun Nam**, Chungryeol Lee, Sukwon Jang, Sanghoon Ahn, Youngsang Cho, "Method for film deposition", (Korea Patent (Priority Claim Country); Application Number: 10-2024-0117056, United States of America Patent; Application Number: 18/921,365).
2. Youngsang Cho, Sung Gap Im, Yongjin Kim, **Taehyun Nam**, Woojin Lee, Chungryeol Lee, Sukwon Jang, "Method for polymeric layer fabrication", (Korea Patent; Application Number: 10-2024-0117056).
3. Sung Gap Im, Sukwon Jang, Chungryeol Lee, **Taehyun Nam**, "Dipole polarization-charge capture synergetic single-polarity driven synapse transistor and artificial sensory system comprising the same", (Korea Patent; Application Number: 10-2024-0193239, United States of America Patent; Application Number: 19/421,173).
4. Sung Gap Im, Juchan Kim, Jeong-ik Park, **Taehyun Nam**, Bohyun Lee, "Polymer thin films for semiconductor devices, semiconductor devices comprising the same, and methods for manufacturing semiconductor devices", (Korean Patent; Application Number: 10-2026-0005661).
5. Sung Gap Im, **Taehyun Nam**, Chungryeol Lee, Minjae Jang, Junyeong Yang, "Radical initiators for chemical vapor deposition and method for producing polymer thin film using the same", (Korean Patent; Application Number: 10-2025-0084923).
6. Sung Gap Im, Wooyoung Chung, **Taehyun Nam**, Changmin Lee, "Sn-based Organic/Inorganic Hybrid UV-sensitive Polymer Thin Film Formed by iCVD Process", (Korean Patent (Priority Claim Country); In submission, United States of America Patent; In submission).
7. Sung Gap Im, **Taehyun Nam**, Chungryeol Lee, Seokhyun Hong, "Gate Injection Synaptic Transistor with Tunable Stochastic Memory for Ensemble Reservoir Computing", (Korean Patent; In submission).